

Sidra Nisar

Data Analyst, Machine Learning and Data Professional Sydney, Australia |
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Data Analytics and Machine Learning professional with a background in Computational Finance and Mathematical Sciences (Data Science). Experienced in data cleaning, statistical analysis, machine learning model development, predictive analytics, and business intelligence reporting across healthcare, financial services, and image analytics domains.

Skilled in transforming large and complex datasets into actionable insights using Python, SQL, Power BI, Excel, R, and machine learning techniques. Experience developing predictive models, performing data validation, conducting statistical testing, and creating analytical dashboards to support data-driven decision-making.

Strong foundation in data preprocessing, feature engineering, visualization, and stakeholder-focused reporting. Passionate about leveraging analytics and emerging AI technologies to solve business problems and improve operational outcomes.

SKILLS

- Data Analytics & Business Intelligence
- Data Cleaning & Preprocessing
- Statistical Analysis & Hypothesis Testing
- Data Visualization & Dashboard Development
- Business Reporting & Performance Monitoring
- Stakeholder Reporting & Documentation
- Machine Learning & Predictive Analytics
- Classification & Regression Models
- Feature Engineering
- Model Validation & Cross Validation
- Natural Language Processing (NLP)
- Deep Learning & Neural Networks
- Programming & Data Tools
- Python (Pandas, NumPy, Scikit-Learn)
- SQL
- R Programming
- Power BI
- Microsoft Excel
- Tableau
- Jupyter Notebook
- Git & GitHub

WORK EXPERIENCE

Data Analytics and Machine Learning Projects

Healthcare Claims Predictive Analysis

2025

Tasks

- Analyzed healthcare claims datasets containing over 500,000 records to identify patterns and improve predictive performance.
- Designed and evaluated multiple machine learning models for classification of imbalanced healthcare outcomes.
- Implemented CatBoost models utilizing native categorical feature handling, achieving a 13% improvement in precision-recall Performance.

- Conducted statistical validation through k-fold cross-validation and paired t-tests to ensure model reliability.
- Produced technical documentation and analytical reports to communicate findings and recommendations.

Bank Market Analytics

University of Wollongong

2025

Tasks

- Developed predictive models to analyze customer subscription behaviour using banking marketing datasets.
- Performed end-to-end data preparation including missing value treatment, feature engineering, categorical encoding, and class balancing.
- Applied Logistic Regression, Random Forest, and Support Vector Machine models with hyperparameter optimization.
- Generated business insights to support targeted marketing strategies and customer acquisition decisions.
- Evaluated model performance using precision, recall, F1-score, and cross-validation techniques.

Image Analytics Project

University of Wollongong

2025

Tasks

- Processed and analyzed more than 10,000 images across 1,000 categories using ImageNet datasets.
- Extracted deep-learning feature embeddings using EfficientFormerV2 architecture.
- Improved analytical processing efficiency by approximately 8x through GPU-enabled workflows.
- Developed machine learning classification models achieving approximately 89% predictive accuracy.
- Applied dimensionality reduction and visualization techniques including PCA and t-SNE for model interpretability.
- Delivered end-to-end image analytics pipelines from data acquisition through model evaluation.

Natural Language Processing Chatbot

University of Wollongong

2024

Tasks

- Designed and developed an AI-powered Shakespeare-style conversational chatbot using Python, TensorFlow, PyTorch, and NLP techniques.
- Performed text preprocessing, tokenization, feature extraction, and sequence modelling on large text corpora.
- Built conversational workflows using transformer-based language models and retrieval-augmented methodologies.
- Implemented FAISS indexing and Streamlit interfaces to improve response retrieval and user interaction.
- Conducted model evaluation and optimization to enhance conversational accuracy and system performance.

Quantum Retail Analysis

Virtual Internship Quantum

2023

Tasks

- Cleaned and analysed 250,000+ retail transactions using R.
- Identified key customer segments and purchasing behaviours through data analysis.
- Delivered actionable recommendations based on sales and store performance insights.

Junior Accountant

Sunshines cable, Karachi, Pakistan.

2020

Tasks

- Processed accounts payable and accounts receivable transactions, ensuring accurate and timely recording of financial data.
- Assisted in preparing monthly financial reports, bank reconciliations, and account reconciliations.
- Maintained financial records and supported the month-end closing process.
- Verified invoices, purchase orders, and expense claims for accuracy and compliance.
- Prepared and updated spreadsheets using Microsoft Excel for financial analysis and reporting.
- Monitored cash flow transactions and maintained accurate bookkeeping records.

EDUCATION

- **Master of Mathematical sciences** – University of Wollongong (UOW) | 2025
- **Bachelor of Computational Finance** – NED University of Engineering & Technology | 2017

VISA INFORMATION

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